CODSOFT Python Programming Internship

Task 2: Calculator Application

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- Internship Domain:

  Python Programming

Project

Calculator Application Using Python

Objective

The goal of this project is to create a Calculator Application in Python. This application will let users do math operations. It has a menu that users can interact with. It also keeps track of what calculations were done

Problem Statement

Doing math by hand can take a lot of time. It can also lead to mistakes. A calculator application helps users do math fast and correctly.

The application must:

* Add numbers

* Subtract numbers

* Multiply numbers

* Divide numbers

* Keep track of calculations

* Handle inputs

* Prevent dividing by zero

Tools and Technologies Used

Tool	Purpose
Python 3	Programming Language
VS Code / IDLE	Code Editor
Command Prompt / Terminal	Program Execution

System Requirements

- **Hardware Requirements**
  * Computer/Laptop
  * Minimum 2 GB RAM

- **Software Requirements**
  * Python 3.x
  * Any Code Editor (VS Code, PyCharm, IDLE)

Algorithm

- **Step 1**

Make a list called `History`. This list will store what calculations were done.

- **Step 2**

Show the calculator menu.

- **Step 3**

Ask the user to choose an operation.

- **Step 4**

If the user chooses a math operation:

 * Ask for two numbers.

 * Do the chosen operation.

 * Show the result.

 * Save the operation in history.

- **Step 5**

If the user chooses to view history:

 * Show all calculations.

- **Step 6**

If the user chooses to exit:

 * Close the application.

Source Code

```
# Calculator Application with History


History = []
```

```
while True:
    print("\n===== Calculator =====")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication")
    print("4. Division")
    print("5. View History")
    print("6. Exit")

    choice = input("Enter your choice (1-6): ")

    if choice == '6':
        print("Calculator Closed.")
        break

    elif choice == '5':
        print("\n===== Calculation History =====")

        if History:
            for i, (num1, num2, operation, result) in enumerate(History, start=1):
                print(f"{i}. {num1} {operation} {num2} = {result}")
        else:
            print("No calculations performed yet.")

    elif choice in ['1', '2', '3', '4']:
        try:
            n1 = float(input("Enter First Number: "))
```

```
    n2 = float(input("Enter Second Number: "))

    if choice == '1':
        result = n1 + n2
        print("Result of Addition =", result)
        History.append((n1, n2, "+", result))

    elif choice == '2':
        result = n1 - n2
        print("Result of Subtraction =", result)
        History.append((n1, n2, "-", result))

    elif choice == '3':
        result = n1 * n2
        print("Result of Multiplication =", result)
        History.append((n1, n2, "*", result))

    elif choice == '4':
        if n2 == 0:
            print("Error! Division by zero is not allowed.")
        else:
            result = n1 / n2
            print("Result of Division =", result)
            History.append((n1, n2, "/", result))

except ValueError:
    print("Please enter valid numbers.")
```

```
else:
    print("Invalid Choice! Please select between 1 and 6.")
```

- **Explanation of the Code**

 # History List

Stores all calculations done.

 # Addition

Adds two numbers. Saves the result.

 # Subtraction

Subtracts one number from another.

 # Multiplication

Multiplies two numbers.

 # Division

Divides one number by another. It also prevents dividing by zero.

 # View History

Shows all calculations done before.

 # Exception Handling

Handles inputs using `try-except`.

- **Sample Output**

===== Calculator =====

1. Addition
2. Subtraction
3. Multiplication

4. Division

5. View History

6. Exit

Enter your choice (1-6): 1

Enter First Number: 20.2

Enter Second Number: 60.8

Result of Addition = 81.0

- **History Output**

===== Calculation History =====

1. 20.2 + 60.8 = 81.0

2. 50.0. - 10.0 = 40.0

3. 10.0 * 5.0 = 50.0

4. 100.0 / 4.0 = 25.0

 # Features

- Addition

-Subtraction

-Multiplication

- Division

- Calculation History

- Error Handling

- Division by Zero Protection

- Menu Driven Interface

Advantages

* Easy to use.

* Does calculations fast.

* Stores calculations.

* Handles errors well.

* Good, for beginners.

Learning Outcomes

This project helped learn:

* Python Basics

* Loops

* Conditional Statements

* Handling User Input

* Lists

* Arithmetic Operations

* Exception Handling

* Menu Driven Programming

Future Enhancements

* Scientific Calculator

* Square Root Function

* Modulus Operation

* Power Calculation

* Save History to File

#Conclusion

The Calculator Application was made and works. It does math operations. It keeps track of calculations. Handles errors. This project shows how to use Python for applications.

 # Result

The Calculator Application works. Users can do math operations. They can view calculations and exit the application. The application is easy to use.